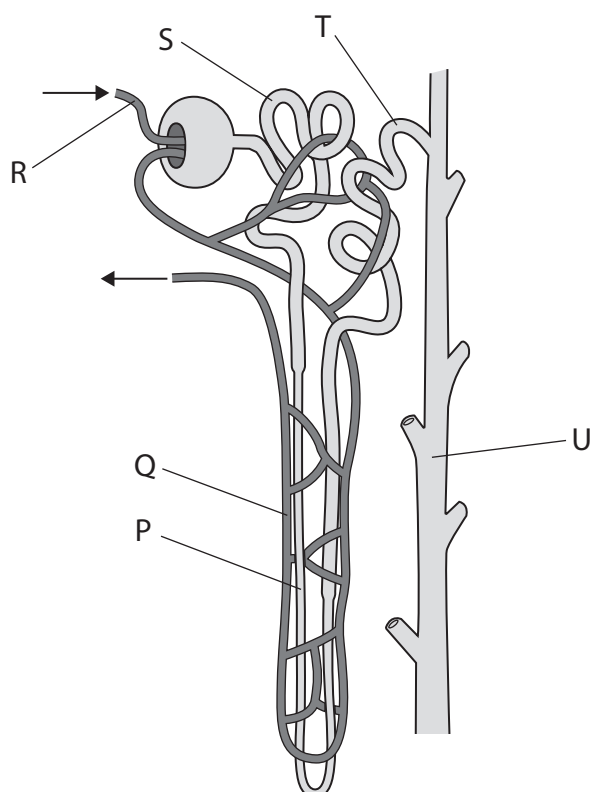


7 The kidney is involved in the reabsorption of solutes and water.

(a) The diagram shows a mammalian kidney tubule (nephron) with its blood supply.



(i) Which labelled structure transports blood to the renal vein?

(1)

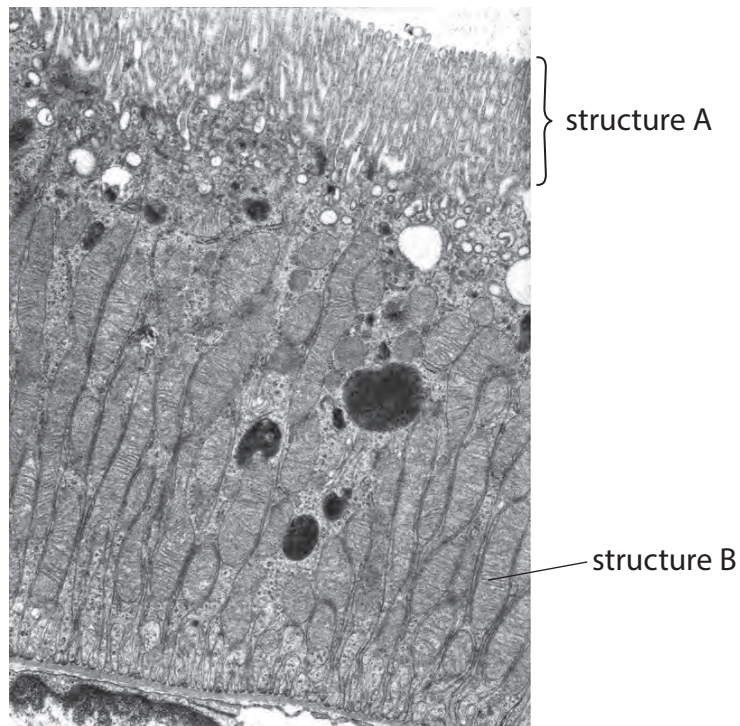
- ☐ **A** Q
- ☐ **B** R
- ☐ **C** S
- ☐ **D** U

(ii) Which labelled structures would be found in the medulla region of the kidney?

(1)

Structures			
☐ A	P	Q	U
☐ B	R	Q	P
☐ C	R	S	T
☐ D	S	T	U

- (iii) The electron micrograph shows part of a cell lining the proximal convoluted tubule.



(Source: © Jose Calvo/Science Photo Library)

Explain how structures A and B and their locations in the proximal convoluted tubule enable the reabsorption of solutes.

(3)

[illegible]

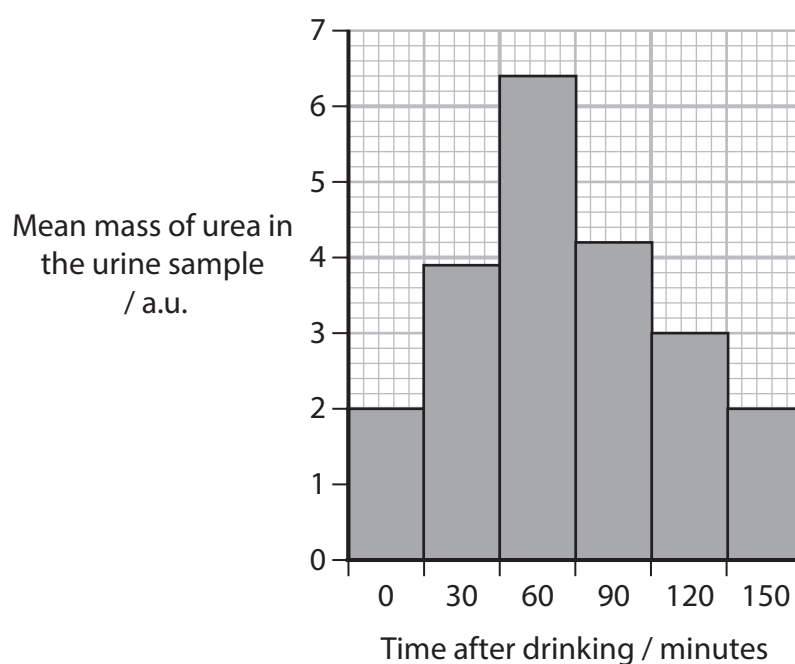
- (b) A group of individuals gave a urine sample and then immediately drank $1\,000\text{ cm}^3$ of water.

Urine samples were then collected at 30-minute intervals.

The volumes and composition of urine produced were then recorded.

The table and graph show the results of this investigation.

Time after drinking / minutes	Mean volume of urine produced / cm^3	Mean concentration of ions in urine / a.u.
0	100	6.0
30	195	3.0
60	325	1.8
90	210	2.8
120	150	4.0
150	100	6.0



Comment on the results of this investigation.

Use the information in the table and the graph and your own knowledge to support your answer.

(4)

[illegible]

(c) A person with hyponatraemia suffers from excessive production of antidiuretic hormone (ADH) from the pituitary gland.

This causes changes in the sodium ion concentration of the blood.

The table shows the sodium ion concentration of the blood in patients with hyponatraemia.

Condition	Sodium ion concentration of the blood / a.u.
normal	135 to 145
mild hyponatraemia	130 to 134
moderate hyponatraemia	125 to 129
severe hyponatraemia	<125

Explain how the excessive release of antidiuretic hormone (ADH) causes these changes in blood sodium concentrations.

(3)

(Total for Question 7 = 12 marks)